



Representation of lattices via set-colored posets

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ABSTRACT

This paper proposes a representation theory for any finite lattice via set-colored posets, in the spirit of Birkhoff for distributive lattices. The notion of colored posets was introduced in Nourine (2000) [34] and the generalization to set-colored posets was given in Nourine (2000) [35]. In this paper, we give a characterization of set-colored posets for general lattices, and show that set-colored posets capture the order induced by join-irreducible elements of a lattice as Birkhoff's representation does for distributive lattices. We also give a classification for some lattices according to the coloring property of their set-colored representation including upper locally distributive, upper locally distributive, meet-extremal and semidistributive lattices.

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1. Introduction

All the objects considered are finite. This paper is motivated by representation theory and its algorithmic consequences for combinatorial objects structured as finite lattices. Whenever you are familiar with Birkhoff's theorem, the intuition behind this new representation is the following: *Take a poset, say $P = (X, \leq)$, a set of colors M and color the elements of P by subsets of M . Then the set of all ideal colors sets of P has a lattice structure and every lattice can be obtained in this way. For example if each element of P has exactly one color then the obtained family of ideal colors sets corresponds to an antimatroid. Moreover, if any two elements have different colors, then the lattice is distributive.*

The question “Why develop lattice theory?” was considered by Birkhoff [3,5] and extended by Wille [39,14] using Formal Concept Analysis (FCA).

The purpose of this paper is to present a representation of lattices that allows us to understand lattice theory. Let us first recall the famous Birkhoff's representation theorem for distributive lattices. All results in this paper can be stated for the dual, by replacing join-irreducible with meet-irreducible elements.

Theorem 1 ([4]). Any distributive lattice L is isomorphic to the lattice of all order ideals $\mathcal{I}(J(L))$, where $J(L)$ is the poset induced by the set of join-irreducible elements of L .

Birkhoff's theorem has been widely used to derive algorithms in many areas. In fact, whenever a set of objects has a structure isomorphic to a distributive lattice, then there exists a poset where the set of its order ideals is isomorphic to the lattice, e.g. stable marriages [18], stable allocation [2], minimum cuts in a network [22], etc.

For general lattices, the unique well known representation is based on sets [6] or a binary relation between join-irreducible and meet-irreducible elements, called the bipartite irreducible poset $Bip(L) = (J(L), M(L), \preceq_L)$ by Markowsky

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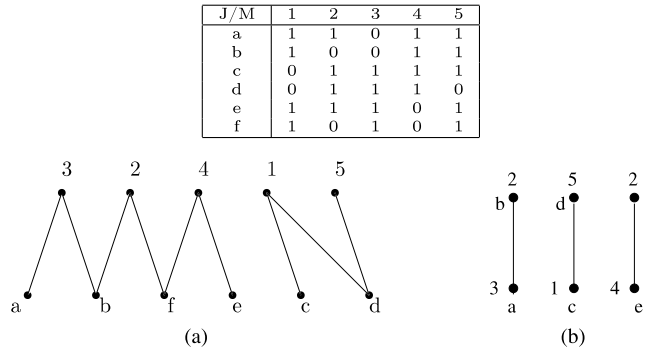


Fig. 1. The table shows a context or binary relation: (a) its corresponding bipartite irreducible poset, and (b) its corresponding set-colored poset.

[30–32] or a context $B(L) = (J(L), M(L), \leq_L)$ by Wille in the framework of Formal Concept Analysis (FCA) [14,39]. From the FCA perspective, elements in $J(L)$ are interpreted as objects and those of $M(L)$ are understood as properties or attributes characterizing these objects. Several other lattice representations have been proposed such as closure systems, implicational systems, and join-core, but all of them use exponential size (see [11,9,24,28]). Lattices representation, FCA and its algorithmic aspects are essential topics in data analysis, as they aim at identifying knowledges and restructuring them as a hierarchy (the reader is referred for examples to the ICFA series of conferences). Over the past two decades, many algorithms have been introduced to consider the reconstruction of a lattice from its representation. Fortunately there exist linear time reconstruction algorithms for distributive lattice (see for example [19,23,38]). Enumeration or reconstruction algorithms for general lattices are equivalent to enumerate maximal bicliques of a bipartite graph (see [15] for a detailed analysis).

Compared to the representation of distributive lattices, the bipartite irreducible poset or the context does not take into account the order inherited by join-irreducible or meet-irreducible elements and the fact that the elements of the lattice correspond to some order ideals of the poset $J(L) = (J(L), \leq)$. In this sense the proposed algorithms for the general case have a bad behavior whenever the lattice is distributive or near to be distributive (e.g. upper locally distributive, semidistributive, ...).

In this paper, we derive a new representation for general lattices via set-colored posets which generalizes the notion of colored posets for representing upper locally distributive lattice in [34,35]. For a lattice L , this representation captures the order inherited by join-irreducible elements $J(L)$ as Birkhoff's representation does for distributive lattices. When restricted to distributive lattices, we obtain an isomorphism between L and the order ideals of $J(L)$. Dilworth [10] has introduced upper locally distributive lattices and has observed that they are close to distributive lattices. Using set-colored posets we confirm this idea and obtain a characterization strongly linked to that of distributive lattices, which can be also deduced from Korte and Lovász [27] and Edelman and Sacks [12] works. Recently, Knauer [26,25] has confirmed this observation using antichains partition. Moreover Magnien et al. [29] have shown that configurations of a Chip-Firing game are structured as upper locally distributive lattice (see Kolja [26]).

Fig. 1 shows an example of a set-colored poset corresponding to an antimatroid. The interest of this representation is to consider only the poset induced by join-irreducible elements and to color them with some particular meet-irreducible elements. Indeed, the colors of a join-irreducible j is all meet-irreducible m such that $j \not\leq m$ (see later). Clearly this representation uses less space memory.

The remainder of the paper is structured as follows: in Section 2, we introduce the set-colored posets and the lattice of ideal colors sets. Section 3 gives a characterization of set-colored posets which are associated to a lattice. In Section 4, we discuss characterization of particular lattices that are close to distributive lattices.

2. Set-colored posets

In this section we first introduce the notion of set-colored posets and some notations that will be used throughout this paper. For definitions on lattices and ordered sets not given here, see [9,14,37].

A partial order (or poset) on a set X is a binary relation $\leq$ on X which is reflexive, anti-symmetric and transitive, denoted by $P = (X, \leq)$. A set $I \subseteq X$ is said to be an ideal if $x \in I$ and $y \leq x$ implies $y \in I$. For an element $x \in X$ we associate the unique ideal $\downarrow x = \{y \in X \mid y \leq x\}$. The set of all ideals of P is denoted by $\mathcal{I}(P)$.

Let $A \subseteq X$, an element $z \in X$ is an upper bound of A if $x \leq z$ for any $x \in A$. The element z is the least upper bound if $z \leq z'$ for all upper bounds z' of A . Dually, we define the greatest lower bound. A partial order $L = (X, \leq)$ is called a lattice if for every two elements $x, y \in X$ both the least upper bound and the greatest lower bound exist, denoted by $x \vee y$ and $x \wedge y$.

Let $L = (X, \leq)$ be a lattice. The element $z \in X$ is a join-irreducible (resp. meet-irreducible) if $z = x \vee y$ (resp. $z = x \wedge y$) implies $z = x$ or $z = y$. The set of all join-irreducible (resp. meet-irreducible) elements of L is denoted by $J(L)$ (resp. $M(L)$).

Let L be a finite lattice and $x, y \in L$. We will use the following arrow relations [14], that are weakening of the so called perspectivity relations defined in lattices (see [17]): $x \not\leq y$ means that x is a minimal element of $\{z \in L \mid z \not\leq y\}$, $x \searrow y$

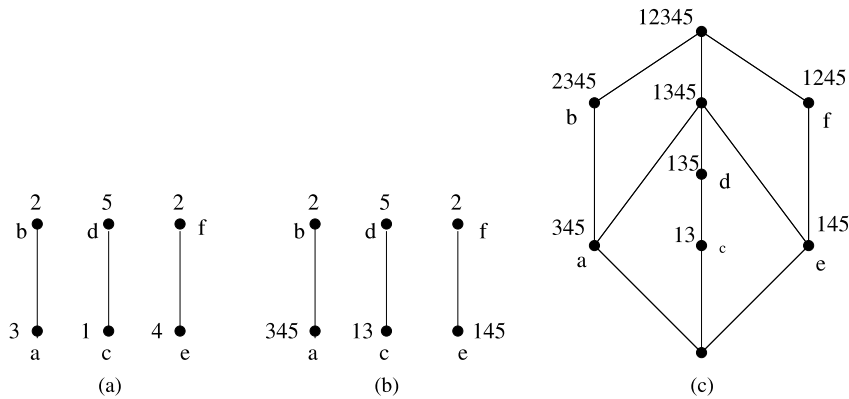


Fig. 2. (a) A proper colored poset, (b) a set-colored poset, and (c) the lattice of ideal colors sets of the set-colored poset in (b).

means that y is a maximal element of $\{z \in L \mid z \not\geq x\}$ and $x \nearrow y$ means that $x \nearrow y$ and $x \searrow y$. Recall that $\nearrow$, $\searrow$, $\nearrow$ are relations defined on $J(L) \times M(L)$.

A set-coloring γ for a poset $P = (X, \leq)$ is a function that assigns a set of colors to every element in X such that for all $x, y \in X$, the sets of colors $\gamma(x)$ and $\gamma(y)$ are disjoint whenever $x < y$. In other words, γ is a set-coloring of the comparability graph of P , as introduced in [36].

Definition 1. A set-colored poset, denoted by $P = (X, \leq, \gamma, M)$, is the poset $(X, \leq)$ equipped with a set coloring $\gamma : X \rightarrow 2^M$ where M is a set of colors. A set-colored poset is said to be a *proper colored poset* if the colors set of any element is a singleton, i.e., the coloring γ is a function from X to M .

Fig. 2(a) and (b) show two examples of set-colored posets.

Let $P = (X, \leq, \gamma, M)$ be a set-colored poset. A subset $C \subseteq M$ is said to be an *ideal colors set* if there exists an ideal I of P such that $C = \gamma(I) = \bigcup_{x \in I} \gamma(x)$. In Fig. 2(b), $C = \{1, 3, 4, 5\}$ is an ideal colors set, since $C = \gamma(\{a, c\})$. Note that two different ideals of P can have the same colors set, i.e. if $\gamma(\{c, e\}) = \gamma(\{a, c\}) = \{1, 3, 4, 5\}$.

The set of all ideal colors sets of P , denoted by $\mathcal{C}(P)$ has a lattice structure as shown in the following:

Proposition 1. Let $P = (X, \leq, \gamma, M)$ be a set-colored poset. Then $\mathcal{C}(P)$ ordered under set-inclusion is a lattice.

Proof. It suffices to show that $\mathcal{C}(P)$ is closed under union and containing the empty set. First we show that $\mathcal{C}(P)$ is closed under union. Let C_1, C_2 be two ideal colors sets of P . Then there exist two ideals I_1 and I_2 such that $\gamma(I_1) = C_1$ and $\gamma(I_2) = C_2$. Since ideals are closed under union, thus $I_1 \cup I_2$ is an ideal and therefore $C_1 \cup C_2$ are its colors. Moreover the ideal colors set corresponding to the empty order ideal is empty. $\square$

Fig. 2(c) shows the lattice of all ideal colors sets of the set-colored poset in Fig. 2(b).

Proposition 2. Let $P = (X, \leq, \gamma, M)$ be a set-colored poset. The mapping $\text{gen} : \mathcal{C}(P) \rightarrow \mathcal{I}(P)$ defined by $\text{gen}(C) = \{x \in X \mid \gamma(\downarrow x) \subseteq C\}$ is an order embedding. Moreover, $\text{gen}(C)$ is the unique largest ideal I of P with $\gamma(I) = C$.

Proof. Let C_1, C_2 be two ideal colors sets of P . We show that $C_1 \subseteq C_2$ iff $\text{gen}(C_1) \subseteq \text{gen}(C_2)$. First suppose $C_1 \subseteq C_2$ and let $x \in \text{gen}(C_1)$. Then $\gamma(\downarrow x) \subseteq C_1 \subseteq C_2$ which implies that $x \in \text{gen}(C_2)$. Now suppose that $C_1 \not\subseteq C_2$. Then there exists $c \in C_1 \setminus C_2$ and $x \in \text{gen}(C_1)$ such that $c \in \gamma(x)$. Then $\gamma(x) \not\subseteq C_2$ and therefore $\text{gen}(C_1) \not\subseteq \text{gen}(C_2)$ since $x \notin \text{gen}(C_2)$.

Assume that there exist two different maximal (under inclusion) ideals I and J with $\gamma(I) = \gamma(J) = C$. Then $\gamma(I \cup J) = C$ since ideals are closed under union, and thus contradicts the fact that I and J are maximal under inclusion with $\gamma(I) = \gamma(J) = C$. $\square$

It is worth noticing that several set-colored posets lead to the same lattice of ideal colors sets. In the following, we give a characterization of such set-colored posets that are (up to isomorphism) canonical, i.e. the elements of X (resp. M) are in bijection with join-irreducible (resp. meet-irreducible) elements of the lattice of ideal colors sets (see Fig. 3).

3. Representing a lattice by a set-colored poset

In this section we show that any lattice L can be represented by a set-colored poset P_L such that its associated lattice $\mathcal{C}(P_L)$ is isomorphic to L .

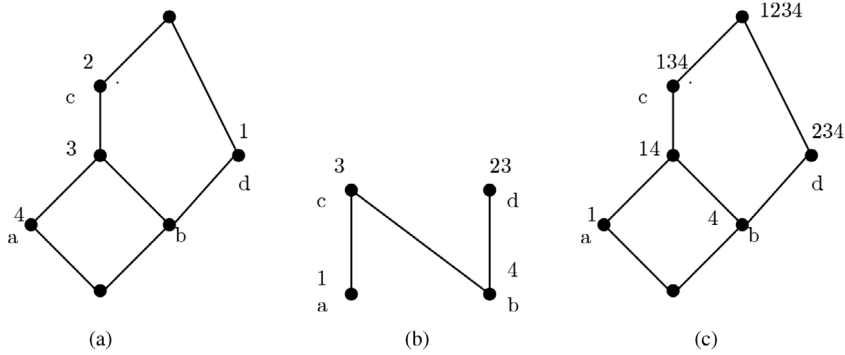


Fig. 3. (a) A lattice L in which join-irreducible (resp. meet-irreducible) elements are labeled with letters (resp. numbers), (b) its associated set-colored poset P_L and (c) the lattice of its ideal colors sets.

Definition 2. Let $L = (X, \leq)$ be a lattice. We denote $P_L = (J(L), \leq, \gamma, M(L))$, the set-colored poset defined by the following set-coloring:

$$\gamma : J(L) \rightarrow 2^{M(L)}, \text{ with } \gamma(j) = \{m \in M(L) \mid j \not\leq m\}$$

It is worth noticing that all results in this paper have their dual. In fact, the dual lattice L^d of L can be represented by $P_{L^d} = (M(L), \geq, \gamma^d, J(L))$, where

$$\gamma^d : M(L) \rightarrow 2^{J(L)}, \text{ with } \gamma^d(m) = \{j \in J(L) \mid j \not\geq m\}$$

Lemma 1. Let $L = (X, \leq)$ be a lattice and the mapping $\varphi : L \rightarrow \mathcal{C}(P_L)$ with $\varphi(a) = \gamma(J(a))$, where $J(a) = \{j \in J(L) \mid j \leq a\}$, then:

1. For every $a \in X$, $\varphi(a) = \{m \in M(L) \mid a \not\leq m\}$
2. For every $a, b \in X$, $a \leq b$ iff $\varphi(a) \subseteq \varphi(b)$, i.e. φ is an order embedding.

Proof.

1. Let $m \in M(L)$ and $a \not\leq m$. Then there exists $j \leq a$ such that $j \not\leq m$. Thus $m \in \gamma(j)$ which implies $m \in \gamma(J(a))$, since $j \in J(a)$. So, $m \in \varphi(a)$.
 Now let $m \in \varphi(a)$. Then there exists $j \in J(a)$ such that $j \not\leq m$. This means that $j \not\leq m$ and then $a \not\leq m$ since $j \leq a$.
2. Let $a \leq b$, for $a, b \in L$. Then $a \not\leq m$ implies $b \not\leq m$. So $\varphi(a) \subseteq \varphi(b)$.
 Now suppose $\varphi(a) \subseteq \varphi(b)$. This means, that for all $m \in M(L)$, $a \not\leq m$ implies $b \not\leq m$. Suppose $a \not\leq b$. Then exists $m \in L \setminus \uparrow a$ such that $a \not\leq m$ and $b \leq m$. This contradicts the fact that $a \not\leq m$ implies $b \not\leq m$. $\square$

We can now formulate our main representation theorem:

Theorem 2. Any lattice L is isomorphic to the lattice of the ideal colors sets of its poset P_L when ordered under inclusion.

Proof. To this aim, let us prove that for a lattice $L = (X, \leq)$, $\varphi : L \rightarrow \mathcal{C}(P_L)$ is an order-isomorphism, with $\varphi(a) = \gamma(J(a))$ and $\varphi^{-1}(C) = \bigvee \text{gen}(C)$.

Using Lemma 1, the mapping φ is an order embedding and therefore one-to-one. It remains to show that φ is onto.

Let $C \in \mathcal{C}(P_L)$ and $a = \bigvee \text{gen}(C)$. Then $a \in L$ since the supremum always exists for a finite lattice. It suffices to show that $\varphi(a) = C$.

Let $m \in C$. Then there exists $j \leq a$ such that $m \in \gamma(j)$, i.e. $j \not\leq m$. Thus $a \not\leq m$ and $m \in \varphi(a)$. So $C \subseteq \varphi(a)$.

Conversely, let $m \in \varphi(a)$. Then there exists $j \leq a$ such that $j \not\leq m$. Suppose $m \notin C$. This implies that $j' \leq m$ for all $j' \in \text{gen}(C)$. Therefore $a = \bigvee \text{gen}(C) \leq m$ which contradicts $a \not\leq m$ and then $m \in \varphi(a)$. So $\varphi(a) \subseteq C$. $\square$

As Birkhoff's Theorem 1 which provides for distributive lattices not only a compact representation via a poset but also some structural insights that can be used algorithmically [19], Theorem 2 does the same for arbitrary lattices. This representation is compact since $|L|$ can be exponential in $|P_L|$.

Before discussing some of the consequences of this result, let us first characterize set-colored posets which are isomorphic to the set-colored poset P_L for some lattice L . Fig. 4 shows three set-coloring of the same poset and none of them is isomorphic to some P_L for a lattice L .

Recall the characterization of those bipartite posets which are isomorphic to $\text{Bip}(L) = (J(L), M(L), \not\leq)$ for some lattice L .

Theorem 3 ([32]). Let $B = (X, Y, E)$ be a bipartite poset. B is isomorphic to $\text{Bip}(L) = (J(L), M(L), \not\leq)$ for some lattice L if and only if the following holds:

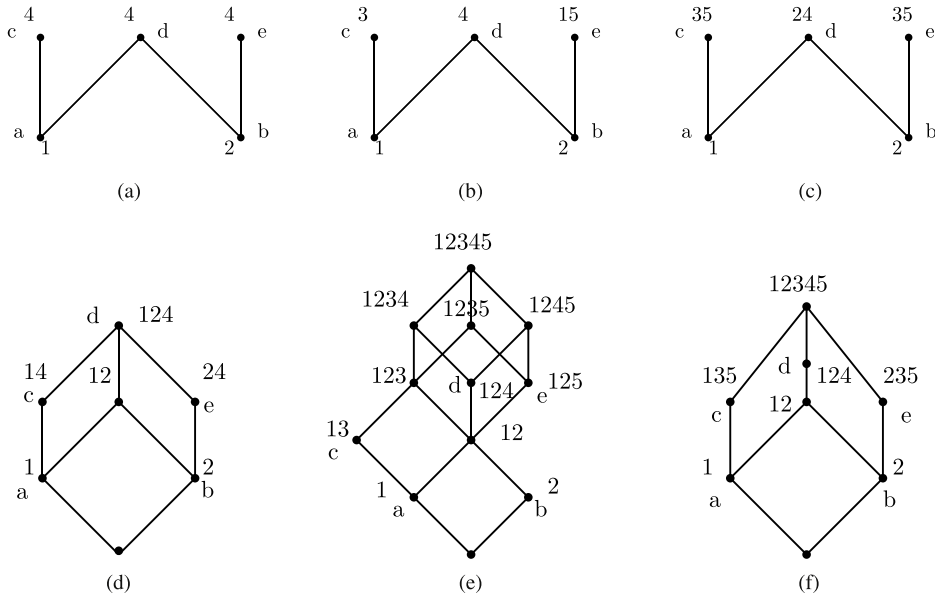


Fig. 4. (d), (e) and (f) are, respectively, the lattices of ideal colors sets of set-colored posets in (a), (b) and (c). In (a) the element d is not a join-irreducible, in (b) the element a is not comparable to e and in (c) the number of colors is greater than the number of meet-irreducible elements.

1. For all $x \in X$, if $W \subseteq X$ is such that $N(x) = \bigcup_{w \in W} N(w)$, then $x \in W$.
2. For all $y \in Y$, if $V \subseteq Y$ is such that $N(y) = \bigcup_{v \in V} N(v)$, then $y \in V$.

where $N(x) = \{y \in X \cup Y \text{ s.t. } xy \in E\}$, i.e. the neighborhood of x in B .

Using the characterization of [Theorem 3](#), we derive a characterization of set-colored posets which are isomorphic to P_L for some lattice L . The following theorem is also a rewriting of the characterization of clarified reduced contexts [\[14\]](#).

Theorem 4. Let $Q = (X, \leq, \gamma, M)$ be a set-colored poset. Then Q is isomorphic to P_L for some lattice L iff the following conditions are satisfied:

1. For all $A \subseteq X$ and $x \in X$, $\gamma(\downarrow x) = \gamma(\downarrow A)$ implies $x \in A$ (**join-irreducible condition**), where $\downarrow x = \{y \in X \mid y \leq x\}$ and $\gamma(\downarrow A) = \bigcup_{a \in \downarrow A} \gamma(a)$.
2. For all $x, y \in X$, $\gamma(\downarrow x) \subseteq \gamma(\downarrow y)$ implies $x \leq y$ (**ordering condition**).
3. For all $m, n, p \in M$, $\beta(m) \neq \beta(n)$ or $\beta(m) = \beta(n) \cap \beta(p)$ implies $m = n$ or $m = p$, where $\beta(m) = \{a \in X \mid m \notin \gamma(\downarrow a)\}$ (**meet-irreducible condition**).

Proof. Suppose Q is isomorphic to some P_L . Since any element $x \in X$ maps to a join-irreducible $j \in J(L)$, then $\gamma(\downarrow x)$ cannot be the join (here the union) of others elements of X . Moreover $(X, \leq)$ is isomorphic to $(J(L), \leq)$ then conditions 1 and 2 hold. Furthermore, every meet-irreducible $m \in M$ maps to a meet-irreducible of $M(L)$ and thus the corresponding ideal $\beta(m)$ cannot be the meet (here intersection) of other elements in M . Thus condition 3 holds.

Conversely, suppose that the three conditions holds. Using join-irreducible and ordering conditions 1, 2 and [Lemma 1](#), the ideal colors set $\gamma(\downarrow x)$ corresponds to a join-irreducible in $\mathcal{C}(Q)$ and thus $(X, \leq)$ is isomorphic to $(J(L), \leq)$ for some lattice L .

Now suppose that condition 3 holds. Clearly, $\beta(m)$ is the maximal ideal such that $m \notin \gamma(\beta(m))$. Thus $\gamma(\beta(m))$ corresponds to a meet-irreducible in $\mathcal{C}(Q)$ since $\beta(m) \neq \beta(n)$ for $m, n \in M$. $\square$

Clearly conditions of [Theorem 4](#) can be checked in polynomial time in the size of the colored poset and thus it can be recognized in polynomial time whether a given set-colored poset is isomorphic to some P_L for a lattice L .

4. Set-colored posets for particular classes of lattices

We show in this section how [Theorem 2](#) unifies many results of lattice theory. First, we derive the famous Birkhoff's representation for distributive lattices, then we show that upper locally distributive lattices correspond to proper colored posets. As a consequence we obtain a characterization of the lattice of all Moore families. We also show that set-colored posets yields nice properties for extremal and semidistributive lattices.

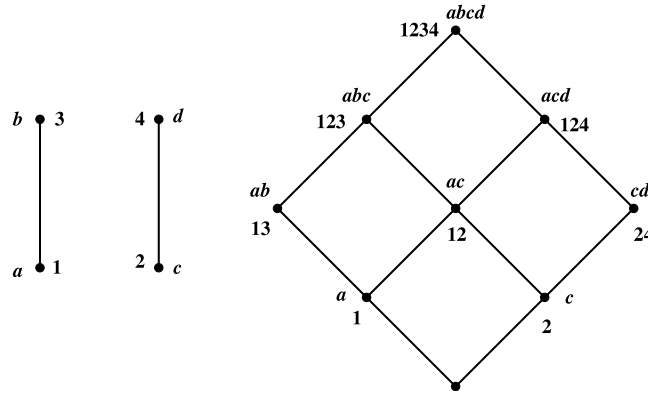


Fig. 5. The set-colored poset of a distributive lattice.

4.1. Distributive lattices

Distributive lattices have been used either theoretically or algorithmically in several areas. In fact several combinatorial objects can be structured as distributive lattices (see for example, Knauer [26] who gives a list of problems from graphs). Fig. 5 shows the colored poset associated to a distributive lattice.

Theorem 5. Let L be a lattice. Then the following statements are equivalent:

1. L is distributive
2. L is isomorphic to the lattice of all ideals of the poset induced by join-irreducible elements of L
3. For each $j \in J(L)$ there exists a unique $m \in M(L)$ such that $j \not\leq m$, and for each $m \in M(L)$ there exists a unique $j \in J(L)$ such that $j \not\leq m$.
4. $P_L = (J(L), \leq, \gamma, M)$ is a proper colored poset and γ is bijective.

Proof. The equivalence between 1 and 2 is due to Birkhoff's Theorem 1. The equivalence between 1 and 3 is due to a Theorem of Wille [14]. To show that 2 is equivalent to 4, it suffices to note that there is a bijection between order ideals of $(J(L), \leq)$ and the ideal colors sets of P_L . Indeed, two different ideals have different colors since γ is bijective. $\square$

4.2. Upper locally distributive lattices

Dilworth [10] has introduced upper locally distributive lattices and has observed that they were close to distributive lattices. Upper locally distributive lattices have been rediscovered many times, and have several names in the literature (such as join-distributive lattices [12], or antimatroids [27], ...). For a survey, [1,33] for other characterizations. Here we will consider an upper locally distributive lattice as an antimatroid. Based on our work [34], Knauer [26] has given an equivalent characterization and gives a list of applications of upper locally distributive lattices.

A lattice is said to be upper locally distributive if for every $x \in L$ the interval $[x, \vee\{y \in L \mid y \text{ covers } x\}]$ is a boolean sublattice of L , where y covers x means that there is no $z \in L$ with $x < z < y$ (see Fig. 6). An antimatroid is a family $\mathcal{F}$ of subsets of a set M such that:

- for every $A, B \in \mathcal{F}$, $A \cup B \in \mathcal{F}$ (closed under union)
- for every non empty set $A \in \mathcal{F}$, there exists $m \in A$ such that $A \setminus \{m\} \in \mathcal{F}$ (Accessibility)

Using the characterization of upper locally distributive lattice by arrows relations, we obtain the following:

Theorem 6. For a lattice L , the following statements are equivalent:

1. L is upper locally distributive.
2. For each $j \in J(L)$ there exists a unique $m \in M(L)$ such that $j \not\leq m$.
3. $P_L = (J(L), \leq, \gamma, M(L))$ is a proper colored poset.
4. $\mathcal{C}(P_L)$ is an antimatroid.

Proof. The equivalence between 1 and 2 is due to Ganter and Wille [14] (dual of Theorem 44), and between 2 and 3 is by definition of P_L . Since $\mathcal{C}(P_L)$ is closed under union, the accessibility is equivalent to proper coloring. $\square$

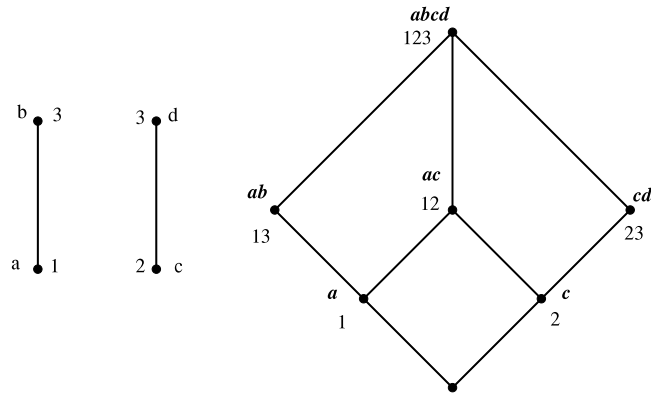


Fig. 6. An example of locally distributive lattice and its associated proper colored poset.

Therefore, distributive and upper locally distributive lattices can be represented by proper colored posets. Furthermore, Theorem 6 confirms Dilworth's observation, since it differs only slightly from Theorem 5. In fact a lattice L is distributive if and only if P_L and P_{L^d} are proper colored posets.

4.3. The lattice of Moore families

In this section we recall the characterization of the lattice of Moore families using proper colored poset [20].

Let X be an n -set and 2^X its power set. A Moore family on X is a family of subsets of X closed under set-intersection and containing the set X . A Moore family is also known as a closure system; i.e. the set of all closed sets of a closure operator. The set $\mathcal{M}_n$ of all possible Moore families on an n -set X , ordered by set-inclusion is a lower locally distributive lattice. By Theorem 6 there exists a colored poset P_n such that the lattice $\mathcal{M}_n$ is dually isomorphic to the set of its ideal colors sets.

Let Q be a boolean lattice on n atoms, say $a_0, a_1, \dots, a_{n-1}$. We consider the mapping $\gamma : Q \rightarrow [0, 2^n - 1]$ as follows:

$$\gamma(x) = \begin{cases} 0 & \text{if } x \text{ is the bottom element} \\ 2^i & \text{if } x = a_i \text{ for some } i \in [0, n-1] \\ \sum_{a \in J(x)} \gamma(a) & \text{otherwise} \end{cases} \quad (1)$$

where $J(x)$ is the set of all atoms below x in Q .

The application γ is a properly coloring since each element of 2^X has only one color. Moreover $x <_Q y$ implies $J(x) \subset J(y)$ and therefore $\gamma(x) \neq \gamma(y)$.

The poset P_n is defined as the disjoint sum of all intervals $[a, \top]$ of Q where a is an atom of Q and $\top$ the top element of Q , i.e. $P_n = \bigcup_{i=0}^{n-1} Q_i$ where Q_i is the induced poset by $[a_i, \top]$ in Q and the coloring is inherited from the coloring of Q (see Fig. 7).

Proposition 3 ([20]). *There is a bijection between the ideal colors sets of P_n and the set of Moore families on an n -set.*

By applying the algorithm to generate ideal colors sets for a colored poset, we found that the number of Moore families on 6 elements is exactly **75.973.751.474** (see [20] for more details). Recently, Colomb et al. [7] have obtained the number of Moore families on $n = 7$ using recursive decomposition of $\mathcal{M}_n$.

Now we derive an explicit characterization of the lattice of all Moore families using the binary relation $R_n = (J_n, M_n, \leq)$ where J_n and M_n are, respectively, the set of join-irreducible and meet-irreducible elements. This is a rewriting of Proposition 3 using binary relation instead of colored poset. Let $J_n = \{a_{ij} \text{ such that } i \in [0, n-1], j \in [0, 2^{n-1} - 1]\}$, $M_n = \{k \in [1, 2^n - 1]\}$ and $a_{ij} \leq k$ iff $(2^i \text{ OR } 2^j) \text{ AND } k = k$ where OR and AND are the logical binary operations.

Corollary 1. *There is a bijection between maximal antichains (bicliques) of $R_n = (J_n, M_n, \leq)$ ($\bar{R}_n = (J_n, M_n, \not\leq)$) and the set of all Moore families on a n -set.*

4.4. Semidistributivity and extremality

In [21] meet-simplicial lattices were introduced. This class of lattices generalizes many known classes of lattices such as meet-extremal and meet-semidistributive lattices.

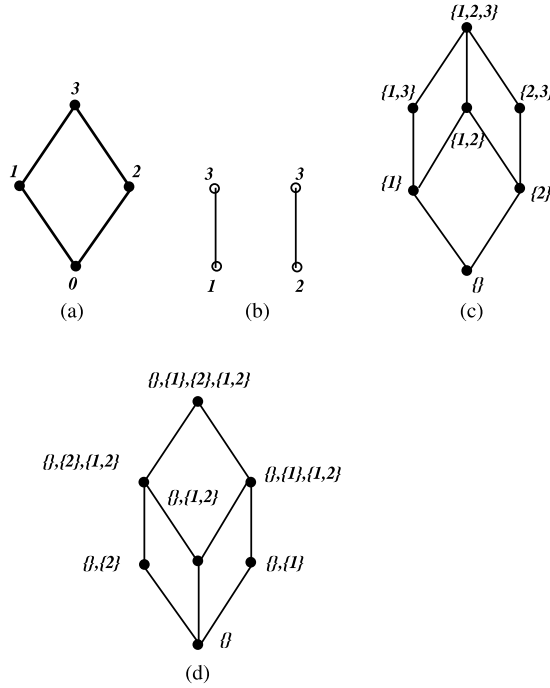


Fig. 7. (a) 2^2 the boolean lattice having two atoms, (b) P_2 the properly colored poset, (c) the ideal colors sets lattice of P_2 , and (d) the lattice of Moore families on the set $X = \{1, 2\}$.

Let L be a lattice and $\sigma = j_1, j_2, \dots, j_{|J(L)|}$ be an ordering of $J(L)$. We denote by $\Delta_i(\sigma)$ the number of colors that do not appear for the first $i-1$ join-irreducibles in P_L , i.e. $\Delta_i(\sigma) = |\gamma(j_i) \setminus \bigcup_{h=1}^{i-1} \gamma(j_h)|$. σ is said to be a linear extension of the induced poset $(J(L), \leq)$ if $i \leq k$ implies $j_i \leq j_k$ in L . We denote by $h(L)$ the size of a maximal chain in L and Δ_i^d for the ordering σ on P_L^d . A lattice L is said to be

- meet-semidistributive (Meet-SD) if for all elements $x, y, z \in L$, $x \wedge y = x \wedge z$ implies $x \wedge y = x \wedge (y \vee z)$.
- join-semidistributive (Join-SD) lattice if for all elements $x, y, z \in L$, $x \vee y = x \vee z$ implies $x \vee y = x \vee (y \wedge z)$.
- semidistributive if it is Meet-SD and Join-SD.
- meet-extremal if $h(L) = |M(L)|$.
- join-extremal if $h(L) = |J(L)|$.
- extremal if it is meet-extremal and join-extremal
- meet-simplicial if there is a total ordering $\sigma = j_1, \dots, j_{|J(L)|}$ such that $\Delta_i(\sigma) \leq 1, i \in [1..|J(L)|]$.
- join-simplicial if there is a total ordering $\sigma = m_1, \dots, m_{|M(L)|}$ such that $\Delta_i^d(\sigma) \leq 1, i \in [1..|M(L)|]$.
- simplicial if it is meet-simplicial and join-simplicial.

The results of Markowsky in [32] on extremality and those of Freese et al. [13] and Ganter and Wille [14] on semidistributivity can now be written in terms of set-colored posets.

Theorem 7 ([32]). *If L is a meet-extremal lattice then there exists a linear extension σ of $(J(L), \leq)$ such that $\Delta_i(\sigma) \leq 1$.*

Corollary 2. *A lattice L is extremal iff there exist two linear extensions σ of the poset $(J(L), \leq)$ and σ^d of $(M(L), \geq)$ such that $\Delta_i(\sigma) \leq 1$ and $\Delta_i^d(\sigma^d) \leq 1$. Furthermore, $\Delta_i(\sigma) = \Delta_i^d(\sigma^d) = 1$.*

Remark 1. Corollary 2 implies that there exists a linear extension σ of $(J(L), \leq)$ such that any join-irreducible j has a proper double arrow ($\nearrow$) that do not appear in the colors of join-irreducible elements preceding j in the linear extension σ .

For an arbitrary lattice, the authors of [13] define the function κ_L that associates for every join-irreducible $j \in J(L)$ $\kappa_L(j) = m \in M(L)$ whenever $\{m\} = \{m' \in M(L) \mid j \nearrow m'\}$, otherwise it is undefined.

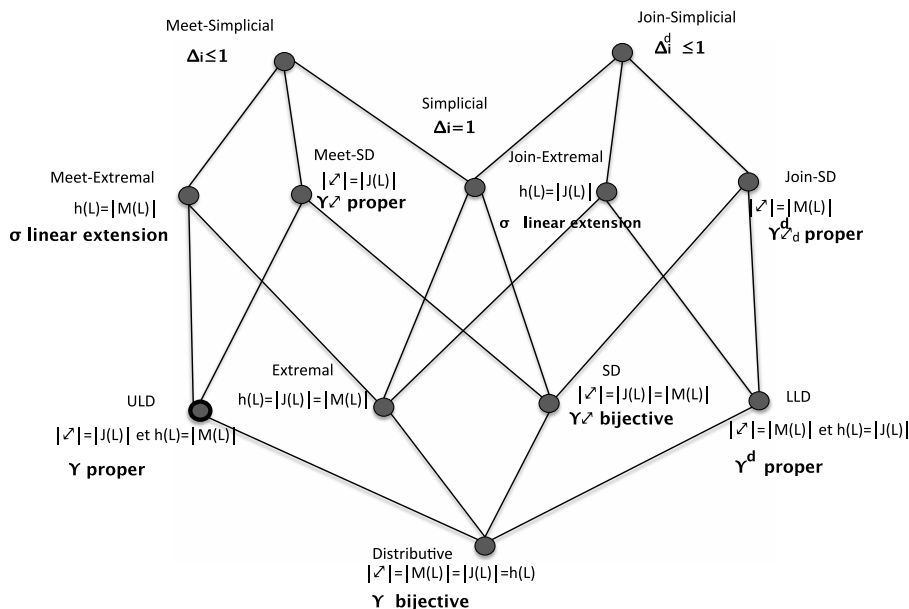


Fig. 8. Hierarchy of lattice classes according to the set-coloring property and simplicial elimination scheme.

Theorem 8 ([13]). Let L be a lattice. The following statements are equivalent:

1. L is meet-SD
2. $\kappa_L(j)$ is defined for every $j \in J(L)$ (or bijection).
3. $|\nabla| = |J(L)| \geq |M(L)|$.

Theorem 9 ([13]). If L is a meet-SD lattice then there exists an ordering σ of $J(L)$ such that $\Delta_i(\sigma) \leq 1$.

Corollary 3. A lattice L is semidistributive iff there exist two orderings σ of the poset $(J(L), \leq)$ and σ^d of $(M(L), \geq)$ such that $\Delta_i(\sigma) \leq 1$ and $\Delta_i^d(\sigma^d) \leq 1$. Furthermore, $\Delta_i(\sigma) = \Delta_i^d(\sigma^d) = 1$.

Remark 2. Theorem 9 and Corollary 3 hold for infinite semidistributive lattice L , whenever L has the minimal join cover refinement property and its dual (see Theorem 2.55 in [13]).

The function κ_L can also be seen as a restriction of the coloring γ of the poset $(J(L), \leq)$ to only double arrows, i.e. $\gamma_{\nabla}(j) = \{m \in M(L) \mid j \nabla m\}$ for every $j \in J(L)$. Using the coloring γ_{∇} : distributive and semidistributive lattices share the same property of coloring. This property of coloring also holds for meet-SD and upper locally distributive lattices, but it is not bijective.

Corollary 4. A lattice L is semidistributive iff $D_L = (J(L), \leq, \gamma_{\nabla}, M)$ is a proper coloring and γ_{∇} is bijective, where $\gamma_{\nabla}(j) = \{m \in M(L) \mid j \nabla m\}$.

Theorem 10. Meet-SD and meet-extremal (resp. SD and extremal) lattices are meet-simplicial (resp. simplicial).

Fig. 8 shows inherited properties of lattices discussed in this paper and their dual.

5. Discussions

Let L be a lattice and $P = (J(L), <, \gamma, M(L))$ its associated colored poset. We define the *chromatic index* by

$$\chi(L) = \max_{j \in J(L)} |\gamma_{\nabla}(j)| = \max_{j \in J(L)} |\{j \nabla m : m \in M(L)\}|$$

We have noticed that upper locally distributive lattices have chromatic index one. A natural problem arises: **Find a characterization of lattices having chromatic index k , $k \geq 2$?**

Geyer [16] has defined k -meet-semidistributive lattices as lattices for which the size of $\gamma_{\nabla}(j) = \{m \in M(L) \mid j \nabla m\}$ is less or equal to k . It is worth noticing that every finite lattice is k -join-semidistributive for some integer k . For lattices of finite length, k -join-semidistributivity can be characterized by a finite list of forbidden sublattices if and only if $k = 1$ [16]. Join-semidistributivity can be characterized by the exclusion of 5 forbidden sublattices [8].

It is well known that lower (upper) bounded lattices are join(meet)-semidistributive. In order to capture properties of bounded lattices, one could enrich the set-colored poset $P = (X, <, \gamma, M)$ with dependencies relations such A, B, C and D that are defined in [13]. Other classes of lattices such as modular or semimodular lattices may be captured using set-colored posets?

We think that this representation of lattices using set-colored posets could be helpful for studying lattice theory and its algorithmic aspects.

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